



Valuing recreational enhancements in the San Patricio Urban Forest of Puerto Rico: A choice experiment approach

Héctor Tavárez^{a,*}, Levan Elbakidze^b

^a Department of Agricultural Economics and Rural Sociology, Agricultural Experiment Station, University of Puerto Rico, Mayaguez, Puerto Rico

^b Resource Economics and Management, Davis College of Agriculture Natural Resources and Design, Center for Innovation in Gas Research and Utilization, West Virginia University, Morgantown, WV, United States

ARTICLE INFO

Keywords:

Choice experiment
Willingness to pay
Urban forest
Recreational enhancements
Puerto Rico

ABSTRACT

Urban forests provide a bundle of benefits, including recreation opportunities. However, the literature on valuation of recreation benefits from urban forests is limited. We use a choice experiment to examine residents' preferences and willingness to pay for different recreation enhancement projects in an urban forest in Puerto Rico. Results from the random parameter logit model show that residents are willing to pay \$29, \$15, \$26 and \$39 for improved trails, an event stage, stands with binoculars and a community garden, respectively. The results suggest that projects aimed at increasing urban forest-based recreation opportunities may be justified conditional on respective project costs.

1. Introduction

Although urban forests provide multiple important ecosystem services,¹ industrialization and urbanization in many countries have reduced the sizes of and access to urban forests. Benefits provided by urban forest ecosystems include improved air and water quality, noise reduction, biodiversity preservation, pollination services and opportunities for recreation (de Groot et al., 2002; Jim and Chen, 2008; Escobedo et al., 2011; Ferraro et al., 2011; Gómez-Baggethun and Barton, 2013; Vujcic and Tomicevic-Dubljevic, 2018). These ecosystem benefits improve residents' quality of life in urban areas (Dwyer et al., 1992; Bolund and Hunhammar, 1999; Chiesura, 2004; Gómez-Baggethun and Barton, 2013) and are important to consider in planning and management of green urban spaces (Ferraro et al., 2011). In particular, it is important to understand the relative values of different services supported by urban forests. In this study, we examine residents' willingness to pay (WTP) for improvements in different recreation opportunities in the San Patricio Urban Forest of San Juan, Puerto Rico. Understanding residents' preferences for different recreation opportunities provided by urban forests supports efficient planning, design and management reflecting relative priorities and benefits generated by different urban forest recreation opportunities.

The literature on valuation of ecosystem services is vast and includes theoretical work (de Groot et al., 2002) and applied valuation of particular services provided by forest ecosystems (Ricketts et al., 2004;

Ferraro et al., 2011; Ninan and Inoue, 2013). Numerous studies estimate values of various forest ecosystem services worldwide, including carbon storage, ecotourism, hydrological services, pollination, health services, and non-timber forest products (Ferraro et al., 2011). The estimated values of forest ecosystem services differ across countries, socio-demographic characteristics, and types of ecosystems (Gürlük, 2006; Brey et al., 2007; Barrio and Loureiro, 2010; Chiabai et al., 2011; Ferraro et al., 2011; Christie et al., 2007; Majumdar et al., 2011; Rosenberger et al., 2012; Upton et al., 2012; Ninan and Inoue, 2013). However, examination of economic values of improvements in different recreational opportunities supported by urban forests in developing countries is limited.

Although WTP values for recreation opportunities provided by forests have been estimated in various locations (Christie et al., 2007; Majumdar et al., 2011; Rosenberger et al., 2012; Upton et al., 2012; Abildtrup et al., 2013; Juutinen et al., 2014; Tyrväinen et al., 2014; Japelj et al., 2016), more research is needed to account for relative priority of different recreation opportunities in urban forests. For example, some residents may prefer passive outdoor recreation opportunities, such as space for meditation or cultural activities, rather than active outdoor recreation like biking trails. Prior studies have demonstrated health benefits of meditation and other cultural activities within urban forest ecosystems (Boncinelli et al., 2015; Vujcic and Tomicevic-Dubljevic, 2018). Community gardens are also important in terms of supporting agricultural education (Saldivar-Tanaka and Krasny, 2004)

* Corresponding author at: Agricultural Experiment Station, 1193 Guayacán Street, Botanical Garden South, San Juan 00926-1118, Puerto Rico.

E-mail addresses: hector.tavarez2@upr.edu (H. Tavárez), levan.elbakidze@mail.wvu.edu (L. Elbakidze).

¹ Ecosystem services are the benefits people obtain from ecosystems (MEA, 2005), including urban forests.

and ecosystem sustainability (Irvine et al., 1999). It is particularly important to understand the relative values of different forest ecosystem service-based recreation opportunities in urban settings where residents often lack access to outdoor recreation. We examine preferences and priorities of residents for several recreation enhancement projects to support urban forest-related planning and management decisions.

A variety of economic valuation methods, including revealed (Ricketts et al., 2004; Pattanayak and Butry, 2005) and stated preference-based methods (Bateman et al., 2002; Birol et al., 2006; Brey et al., 2007; Barrio and Loureiro, 2010; Majumdar et al., 2011; Rosenberger et al., 2012; Johnston et al., 2017) have been used to estimate the values of outdoor recreation and ecosystem services. We use a choice experiment-based survey because it is a convenient method for assessing tradeoffs between multiple attributes. Specifically, this study uses the choice experiment to examine residents' preferences and WTP for improved trails, an event stage, stands with binoculars and a community garden. Choice experiments have been widely used for valuation of non-market goods and services including in the US and the Caribbean (Hanley et al., 1998; Hoyos, 2010; Adams et al., 2011; Schuhmann et al., 2013; Gill et al., 2015; Penn et al., 2016).

The rest of the paper is structured as follows. In section two we describe the study area. Section three describes methodology including survey instrument, theoretical framework, and experimental design. Section four discusses the results, and section five provides concluding remarks.

2. Study area

Puerto Rico is an island located in the Caribbean region with a total estimated population of 3.2 million (US Census, 2018). On September 6, 2017, the island was hit by Hurricane Irma and two weeks later devastated by Hurricane Maria. Due to the hurricanes, two urban forests recognized by the Puerto Rico Department of Natural and Environmental Resources were closed for public visits. One of the forests is the San Patricio Urban Forest, located in San Juan, Puerto Rico. This urban forest has a total surface area of 51.8 acres and is visited mainly by residents of San Juan and Guaynabo, a municipality next to San Juan.

The San Patricio Urban Forest was a US naval base residential area after World War II. At the end of the 1960s the US naval base closed and years later the land was transferred to the government of Puerto Rico (DRNA, 2008). Buildings were demolished between 1977 and 1978, and natural forest was allowed to regenerate (DRNA, 2008). The forest now provides habitat to multiple species, including the Puerto Rican Boa (*Chilabothrus inornatus*). Since 2001 the urban forest has been co-managed by a local committee with a mission to support recreation-related benefits and conservation (CPBSP, 2003). Although the hurricanes have injured the recreation opportunities within the forest, as well as much of the island, residents remain interested in urban forest-based recreation and corresponding forest management policies.

3. Materials and methods

3.1. Focus groups

We used information from various stakeholders to identify four recreation improvement alternatives in the San Patricio Urban Forest. This forest can be developed to accommodate active outdoor recreation opportunities like zip-lining or biking and passive uses like meditation and cultural activities. Two focus groups composed of eleven and six participants, respectively, were convened in the summer of 2017 to select the recreation enhancement projects to be evaluated in the study and to test the survey instrument. Participants from different age groups, income levels, gender and degree of involvement in outdoor recreation were invited to the meetings. The first focus group targeted stakeholders interested in outdoor recreation and forest conservation,

including government officials, university professors and leaders from different non-governmental organizations. The second focus group included residents that live near the forest. The participants filled out the questionnaires, which included Likert-scale and close-ended questions, and were allowed to make comments and convey additional concerns or suggestions. Questions were designed to collect information about preferred recreation activities to develop in the urban forest. The feedback obtained from these focus groups showed that there was significant interest in passive outdoor recreation enhancements including projects to facilitate hiking, bird watching, community gardening, meditation and other cultural activities (educational workshops, small concerts).

Focus groups were also useful for payment vehicle selection in the choice experiment. Residents and community leaders expressed the belief that the access to urban forest should be free. Thus, using a per-visit fee may increase protests responses leading to biased estimates (Bateman et al., 2002; Meyerhoff and Liebe, 2009). The participants also indicated that increasing annual taxes would not be supported by the residents. Although most choice experiment studies in the context of parks or forests ecosystems adopt a per-visit fee as the payment vehicle in each project scenario, we use a one-time payment approach to reduce potential protest answers.

3.2. Survey instrument

The questionnaire consists of three sections. The first section is devoted to the choice experiment designed to estimate WTP for recreation improvement projects. The second section contains questions about the importance of ecosystem services provided by urban forests and choice experiment attributes. The third section includes questions about respondents' socio-demographic characteristics (SDC). This information is used to examine respondents' preferences and choices made in the experiments. The questionnaire was distributed in-person by two interviewers who were trained in interviewing protocol including the potential for interviewer bias.

Each interview progressed as follows: (1) the respondent read a paragraph that explained the purpose of the study and described the proposed outdoor recreation improvement projects, (2) the interviewer carefully explained the choice experiment exercise, (3) the respondent filled out a practice choice experiment table, (4) the respondent answered all choice experiment questions, and (5) the respondent filled out the SDC questions in the last section of the survey. The survey contained a map of the forest (Fig. 1). The map illustrates the urban forest boundaries, trails and proposed locations for the recreation enhancements projects.

3.3. Choice experiment

Choice experiments are commonly used to estimate respondents' WTP for non-market goods and services (Hanley et al., 1998; Hoyos, 2010). In this stated preference-based method respondents are asked to choose between two or more alternatives that differ in one or more attributes (recreational enhancements in this study). Alternatives and attribute levels vary across presented choice tables and respondents. Table 1 shows the attributes, their descriptions and corresponding levels used in the choice experiment.

The set of attributes includes trails, a stage for activities, stands with binoculars and a community garden. The cost attribute represents the price/bid amount for each project scenario. Levels of recreation improvement and cost attributes are selected based on the input received from focus groups participants. \$50 was identified as maximum WTP for recreation opportunities during focus group meetings. An additional level of \$100 is added to capture a potentially higher level that may have been missed during focus groups. Price levels were selected to represent the spectrum of maximum WTP values while keeping the number of levels to four or less. Having too many levels for respondents



Fig. 1. San Patricio Urban Forest and locations of the proposed recreation enhancement projects.

to consider can increase the complexity of the exercise and lead to biased estimates (Hensher, 2006; Hoyos, 2010). An example of a choice set is presented in Table 2.

3.3.1. Experimental design

The choice experiment design includes five attributes, four with two levels and one with four levels. The complete factorial design that includes all possible combinations of attributes and levels includes 64 possible alternatives ($2^4 * 4^1 = 64$). Each choice set has two

alternatives plus a status quo option where all attribute levels reflect the current situation. A complete randomized design would include too many choice sets (Alpizar et al., 2003; Hoyos, 2010). Therefore, an orthogonal fractional factorial design is employed, using Sawtooth software. Although other, more advanced design methods are available, we use the orthogonal fractional factorial design due to its convenience and common use in the experimental economics literature (Richards et al., 2014; Malone and Lusk, 2018; Malone and Lusk, 2019; Wuepper et al., 2019). Orthogonal fractional factorial designs are used to identify

Table 1

Attributes and levels for the choice experiment.

Attributes	Description	Levels
Trails ^a	Managing trails for hiking to enable elderly and people with physical difficulties to walk or move within the forest.	Improved trails Current trails
Stage for activities	Constructing a small stage in a designated space within the forest for different cultural activities, such as small concerts, educational workshops, yoga and meditation classes.	Available Not available*
Stands with binoculars	Installing two stands with binoculars within the forest and some management to support bird watching.	Stands with binoculars No stands or binoculars*
Community garden	Designating space exclusively for community gardening within the forest. The community garden can be used for education purposes and to generate income that can be reinvested in the forest.	Development of a community garden No community garden*
Cost	Household cost of recreation enhancements in each project scenario. Any money paid toward recreation must be considered spent money that cannot be used for other purposes.	\$0* \$10 \$25 \$50 \$100

^a Refers specifically to a new paved trail within the forest (see Fig. 1).

* The attribute level describes the status quo alternative.

Table 2

Example of a choice set used in this study.
Modified from Spanish version.

Recreation	Option A	Option B
Trails	Current trails	Improved trails
Stage for activities	Available	Not available
Stands with binoculars	Stands with binoculars	No stands or binoculars
Community garden	No community garden	Development of a community garden
Cost	\$25.00	\$50.00
Option C None: I would not choose either of these options		

a subset of choice sets without losing relevant information (Louviere et al., 2000). Each respondent is presented with eight choice sets from a predetermined list and is asked to indicate the preferred alternative in each set. The use of multiple choice sets per respondent has been documented in prior literature (Jacobson et al., 2018; Japelj et al., 2016; Olschewski et al., 2012). For example, Rolfe et al. (2000) use 16 choice sets per respondent, while Giergiczny et al. (2015) use 30. We limit the number of choice sets per respondent to eight to avoid the potentially excessive burden, which can affect respondent efforts and data quality.

The survey includes an additional choice set to evaluate respondent comprehension of the valuation exercise. In this choice set, alternative B contains the same recreation enhancement projects as alternative A but at a lower price. Thus, the respondents who understand the exercise should not select alternative A in this choice table. The purpose of including this choice set is to examine the sensitivity of the results to exclusion of data from participants who select the inferior choice.

3.4. Theoretical framework and econometric models

The underlying economic framework for valuation of WTP using choice experiments is based on Lancasterian Consumer Theory. This framework formulates consumer's utility in terms of characteristics of a good rather than the good itself (Lancaster, 1966). This conceptualization enables representation and comparison of alternatives in choice experiments based on the differences in choice attributes. Modeling of respondent decisions in choice experiments is based on Random Utility Theory, which suggests that individual utility is unknown but can be decomposed into deterministic and unobserved components (McFadden, 1974). Respondents analyze and compare the available alternatives and select the one that provides the greatest utility.

The conditional logit and the random parameter logit models are widely used in the analyses of choice experiment data (Hoyos, 2010). In the conditional logit model the utility of an individual i from selecting an alternative j is a function of K attributes, $X_{ij} = \{X_{ij}^1 \dots X_{ij}^K\}$, including the cost, and can be expressed as follows (Sheremet et al., 2018):

$$U_{ij} = ASC_j + \beta'X_{ij} + \varepsilon_{ij} \quad (1)$$

where ASC_j is an alternative-specific constant, β are attribute coefficients, and ε_{ij} is the Independent and Identically Distributed extreme value error. The probability of selecting a specific alternative is defined as:

$$Pr(y_i = j) = \exp(U_{ij}) / \sum_{j=1}^J \exp(U_{ij}) \quad (2)$$

The random parameter logit model extends the conditional logit model by allowing for heterogeneity of individual specific preference parameters and alternative specific constants to vary across participants. Following Sheremet et al. (2018), these parameters are expressed as follows:

$$\beta_{ik} = \beta_k + \delta_k'Z_i + v_{ik} \quad (3)$$

$$ASC_{ij} = ASC_j + \delta_j'Z_i + v_{ij}$$

where β_k is the population mean of the coefficient of k -th attribute assumed to be normally distributed, v_{ik} is the normally distributed respondent specific taste heterogeneity parameter with zero mean, and v_{ij} is normally distributed respondent specific heterogeneity of the alternative-specific constants with zero mean. In this formulation ASC_{ij} and β_{ik} are allowed to vary with respondents' characteristics Z_i , which enter the corresponding equations with coefficients δ_k and δ_j , respectively.

Respondents' WTP for recreational opportunities is given by the negative ratio of the coefficient of the attribute of interest and the cost coefficient (Hoyos, 2010; Greene, 2012). Thus, WTP for attribute k is estimated using the following equation:

$$WTP = -\beta_k / \beta_c \quad (4)$$

where β_k is the coefficient of attribute k and β_c is the cost coefficient.

The conditional logit model is based on the Independence of Irrelevant Alternative (IIA) property, which states that the probability of choosing alternative i over the probability of choosing alternative j is unaffected by the presence or absence of alternative k (Louviere et al., 2000). Although the conditional logit model has some disadvantages relative to the random parameter logit model, such as the assumptions of homogenous preferences across respondents and the IIA (Alpizar et al., 2003; Hoyos, 2010), it is a convenient model for baseline assessment of whether the attributes under consideration are important determinants of choice. Models that relax these assumptions are often preferred to account for unobserved preference heterogeneity across respondents and to improve model fit (Rolfe et al., 2000; Hoyos, 2010).

We follow the Hausman and McFadden (1984) procedure to test whether the IIA assumption holds. The test suggests that if an alternative is irrelevant, excluding it from the model will not change parameter estimates systematically. The test follows a chi-squared distribution with degrees of freedom equal to the number of coefficients to be included in the reduced model and can be denoted as follows:

$$\chi^2 = (\hat{\beta}_s - \hat{\beta}_f)'(\hat{V}_s - \hat{V}_f)^{-1}(\hat{\beta}_s - \hat{\beta}_f) \quad (5)$$

where s denotes the estimators in the restricted subset after removing one alternative, f represents the estimator in the initial model, and $\hat{V}_s$ and $\hat{V}_f$ are the corresponding estimates of the covariance matrices. We carry out the test by dropping each of the recreational enhancement project alternatives and by dropping the status quo alternative from the model (alternative C). The results show that the IIA assumption is not violated when dropping each of the alternatives.²

We estimate three models: a conditional logit model with main effects only, a random parameter logit model with main effects only, and a random parameter logit model that includes SDC of respondents. We use the random parameter logit model to account for unobserved preference heterogeneity across respondents and to obtain coefficients when the IIA assumption is relaxed (Train, 2009; Hoyos, 2010; Domínguez-Torreiro and Soliño, 2011). We incorporate respondents' SDCs in the random parameters logit model to evaluate the effect of respondent characteristics on attitudes toward presented recreation enhancement projects. The respondents' SDCs include household income, respondent education level, age, number of household dependents, involvement in outdoor activities and involvement in environmental protection-oriented institutions, as we expect that these

² The results of the test with six coefficients in the model report a value of 6.75 and 6.71 when alternative A and Alternative B are dropped from the model, respectively. Therefore, we reject the null hypothesis that the IIA property has not been violated, as the statistic exceeds the critical value of the chi-squared distribution with six degrees of freedom (critical chi-squared [6] = 12.59). Similar results are found when the status quo alternative is dropped.

characteristics may lead to different preferences for recreation opportunities. All recreational opportunity attributes are effect-type coded variables (see Champ et al., 2003 for a description of effect-type coded variables) and the cost attribute is defined as a continuous variable in all models. The ASC takes the value of 1 for recreation enhancements alternatives and zero for the status quo option.

3.5. Data

Two hundred eighty-one respondents completed the questionnaire from April to July 2018.³ A total of 22 respondents did not adequately understand the choice experiment exercise as they selected a dominated alternative in the test choice set. The average age for all respondents is forty-eight (Table 3). Sixteen percent of respondents have studied or worked in environmental protection-oriented institutions. The average number of dependents is 1.2 persons per household.

Fifty-four percent of the residents in this study have a bachelor's degree, which is higher than the proportion of bachelor's degree holders (21%) in the municipality (Puerto Rico Planning Board, 2017). Mean household income for the population of San Juan is \$3617/month. According to the results from Likert-scale questions the sample average household income is \$1500–\$3000/month. Forty-five percent of the population in the municipality are male, compared to 37% in our sample. The median age of respondent for the sample is 47, which is slightly higher than the median age of 42 for the population.

4. Results and discussion

Respondents were asked to rank choice experiment attributes from the most important to the least important (1 = most important, 5 = least important). Results from the closed-ended questions suggest that the two most preferred recreation enhancement projects are improved trails (Mean = 2.10, SD = 1.23) and community garden (Mean = 2.41, SD = 1.18), and the two least important attributes are stands with binoculars (Mean = 3.47, SD = 1.15) and stage for activities (Mean = 3.67, SD = 1.20). In addition, respondents were allowed to make comments about the recreational enhancements. Six respondents indicated that cultural activities in the forest may be problematic due to potential noise. Three respondents stated that trails should not be modified for hiking because this would alter the ecosystem. Two respondents mentioned that other recreational opportunities may be prioritized over bird-watching. None of the respondents expressed concerns about developing a community garden in the urban forest. Ten respondents mentioned that community gardens contribute to public engagement with the urban forests. This preliminary information suggests that community garden and improved trails may be among the most preferred recreational enhancement projects.

The status quo alternative is selected 241 times out of a total of 2248 choice sets representing 11% of respondents' choices, which is a reasonable outcome for choice experiments. Previous studies on forest valuation have reported similar (Tyrväinen et al., 2014) or higher rates of status quo selection (Meyerhoff and Liebe, 2009; Czajkowski et al., 2016). At the end of the choice experiment the respondents who select the status quo option in at least one choice set are asked to identify the reasons for why they chose not to support the presented recreation improvement projects.⁴ Respondents can identify more than one reason

for selecting the status quo from presented list options.

Reasons for selecting the status quo options can be grouped into valid or protest motivated. Following Bateman et al. (2002), status quo choices made because of budget limitations, preference for substitute goods, or lack of interest in urban forest-based recreation are considered valid. On the other hand, status quo choices are considered to be protest motivated if the choices are driven by the resistance to paying more taxes in general, disagreement with the measurement unit, preference for government provision of the service, and skepticism about how the money will be used. Protest answers reflect disapproval of the provision mechanism or institutional arrangement rather than lack of any value for the recreation improvement project (Bateman et al., 2002; Meyerhoff and Liebe, 2009). Table 4 provides a summary of the identified status quo selection reasons. Most reasons for selecting the status quo were considered valid. However, eighteen respondents indicated protest motives as the reason for selecting the status quo.

We conduct sensitivity analysis to compare the results of the estimation using all observations with the results of the estimation excluding protest responses and responses of participants who selected the dominated alternative in the test choice table. The model that includes all observations has a better model fit based on the Akaike Information Criterion (AIC). Based on the non-overlapping confidence interval method (Park et al., 1991), there is no statistical differences in WTP estimates across the two scenarios. Therefore, since there is no clear agreement on best practices about inclusion of data from protest responses (Johnston et al., 2017), we present the results from estimation that uses all observations.

All attribute coefficients in the conditional and random parameter logit models are statistically significant (Table 5). The coefficients of all recreational enhancements are positive, suggesting that improvements in recreational opportunities increase the probability of selecting an alternative. The cost coefficient is negative, indicating that the higher the cost, the lower the support for the recreation improvement project. The coefficient of the ASC is negative and significant, indicating a preference for the status quo. The negative sign of the ASC is often interpreted as a status quo bias or endowment effect in choice experiments (Hoyos, 2010). The standard deviation of improved trails, stands with binoculars and community garden are significant in the random parameter logit models, suggesting preference heterogeneity across respondents for these recreational activities. Respondents with higher incomes and education are more likely to support recreational enhancements, and respondents who have environmental protection-oriented education or work experience are less likely to support the projects. Other SDCs listed in Table 3 are not statistically significant. Results from random parameters logit model with SDC of individuals have the best fit in terms of AIC.

Two factors may explain the negative effect of involvement in environmental protection-oriented institutions on the support for recreation improvement projects. First, developing recreational opportunities in the urban forest implies a greater disturbance of the ecosystem. Therefore, respondents with environmental conservation priorities may not favor such projects. Second, respondents who prioritize environmental protection may prefer outdoor recreation in relatively more undisturbed forests. In this respect enhancing recreation opportunities like adding a stage or binocular stands, maintaining hiking trails, or adding community gardens can be undesirable.

Education variable has a positive effect on respondent choices. Thus, the residents with higher level of education are more willing to support recreation enhancement projects than other residents. It is important to note that our sample of residents is generally more educated than the average of the population in San Juan. Although differences between the study sample and population moments have been observed in prior stated preference-based analysis (Domínguez-Torreiro and Soliño, 2011), it is important to consider this caveat for proper interpretation or future studies.

WTP for recreation improvement projects is estimated as the

³ Although financial constraint did not allow us to interview greater number of participants, the number of participants in this study is within the range of sample sizes commonly observed in choice experiment studies.

⁴ None of respondents selected the status quo option in all 8 choice sets. However, it is possible that respondents may protest after completing several choice sets. Therefore, for the sake of robustness check, we obtain estimates using the full data set as well as excluding the responses of participants whose status quo selections may have been protest motivated. The results are not sensitive to exclusion of protest choices.

Table 3
Socio-economic and demographic characteristics of individuals^a.

Variables	Description	Mean (SD)
Gender	Gender of respondent (1 = male, 0 = female)	0.37 (0.48)
Age	Age of respondent	47.59 (16.56)
Income	Total household income per month (1 = less than or equal to \$500, 7 = more than \$7000)	3.32 (1.24)
Education ^b	Education of respondents (1 = none, 5 = graduate school)	4.05 (0.73)
Environment	Respondents who have studied or worked in environmental protection-oriented institutions (1 = yes, 0 = no)	0.16 (0.36)
Number of dependents	Number of household dependents	1.15 (1.24)

SD = Standard deviation.

^a According to the U.S. Census Bureau and Puerto Rico Planning Board, 21% of residents in San Juan have a bachelor's degree, median age is 42, 45% of residents are male and mean household income is \$42,098.

^b The description of categories for education is as follows: 1 = None, 2 = Primary school, 3 = High school, 4 = University, 5 = Graduate school.

Table 4
Reasons for selecting the current situation (status quo).

Reasons	Valid (✓) vs. protest (X)	Frequency ^a (out of 241)	Percent
I am not interested in urban forests	✓	1	0.4
I don't have money to pay for the projects	✓	19	7.9
The prices are too high	✓	103	42.7
The recreational activity of my interest is not available	✓	152	63.1
I prefer to spend my money in other ways	✓	7	2.9
I don't know how my money will be used	X	13	5.4
The government should pay for the projects	X	18	7.5
I don't agree with paying more taxes	X	13	5.4
Neither the forest nor its benefits should be valued in monetary units	X	15	5.5
I didn't understand the options or the selections	X	7	2.9

^a Respondents were allowed to select more than one reason.

Table 5
Results from conditional and random parameter logit models.

Variables	Conditional logit	Random parameter logit	Random parameter logit with SDC
		Random parameters	Random parameters
Improved trails	0.483 (0.033)***	0.570 (0.062)***	0.572 (0.062)***
Stage for activities	0.252 (0.032)***	0.297 (0.045)***	0.301 (0.046)***
Stands with binoculars	0.409 (0.033)***	0.484 (0.057)***	0.504 (0.058)***
Community garden	0.618 (0.035)***	0.732 (0.074)***	0.759 (0.076)***
ASC	0.299 (0.111)***	0.255 (0.124)***	-1.189 (0.436)***
		Fixed parameters	Fixed parameters
Cost	-0.016 (0.001)***	-0.019 (0.002)***	-0.020 (0.002)***
Income	-	-	0.197 (0.070)***
Environment	-	-	-0.687 (0.201)***
Education	-	-	0.236 (0.115)**
		Standard deviation	Standard deviation
Improved trails	-	-0.401 (0.163)***	-0.332 (0.166)**
Stage for activities	-	0.047 (0.451)	0.218 (0.189)
Stands with binoculars	-	0.410 (0.163)***	0.448 (0.148)***
Community garden	-	0.347 (0.195)*	0.416 (0.169)***
ASC	-	-0.083 (0.415)	-0.131 (0.529)
Observations	6744	6744	6600
Respondents	281	281	275
AIC	3385.93	3390.97	3294.95

ASC - Alternative-Specific Constant. SDC of respondents are interacted with the ASC.

Standard Error in parentheses.

* Significant at 0.10.

** Significant at 0.05.

*** Significant at 0.01.

Table 6
Willingness to pay for recreational enhancements across models.

Recreational opportunities	Conditional logit	Random parameter logit	Random parameter logit with SDC
Improved trails	\$29.67 (24.57–34.77)	\$30.03 (24.79–35.26)	\$29.05 (23.92–34.18)
Stage for activities	\$15.50 (11.31–19.70)	\$15.66 (11.42–19.90)	\$15.27 (11.04–19.51)
Stands with binoculars	\$25.15 (20.33–29.97)	\$25.53 (20.57–30.48)	\$25.58 (20.63–30.53)
Community garden	\$37.95 (32.23–43.68)	\$38.57 (32.64–44.50)	\$38.50 (32.59–44.41)

95% confidence intervals are provided in parenthesis.

negative ratio of the regression coefficient for the attribute of interest and the cost coefficient (eq. 4). Results from random parameter logit model with participants' SDC, which has the best fit, indicate that the respondents are willing to pay \$29.05, \$15.27, \$25.58 and \$38.50 for improved trails, a stage for activities, stands with binoculars, and a community garden, respectively (Table 6). WTP estimates for recreation enhancement projects are not statistically different across models based on the non-overlapping confidence interval method (Park et al., 1991).

While most studies estimate WTP values for recreation on a per-visit basis, our estimates are obtained in terms of a one-time payment. Hence, our WTP values are greater than the values estimated on per-visit or annual payment basis. For example, Termansen et al. (2013) estimate that Danish residents are willing to pay approximately \$4 per visit to forested sites. Brey et al. (2007) estimate that Catalan citizens in Spain are willing to pay approximately \$7 per year for forest picnicking activities. At 5% discount rate this corresponds to a one-time WTP of \$140, which is greater than the WTP estimates for recreation enhancement projects we obtain in the current study. This is not surprising given relatively low-income in Puerto Rico and the hardship inflicted by the Hurricanes in 2017.

Consistent with prior choice experiment studies, we find that the

Table 7
Results from logistic regression for scope test.

Variables	Coefficient	Marginal effects
Number of recreation improvements		
One improvement	1.498 (0.105)***	0.344 (0.024)***
Two improvements	2.664 (0.092)***	0.575 (0.016)***
Three improvements	3.436 (0.105)***	0.691 (0.013)***
Four improvements	4.425 (0.176)***	0.718 (0.009)***
Cost	−0.016 (0.001)***	−0.003 (0.000)***
Constant	−2.112 (0.065)***	
N	6744	
Pseudo R-Squared	0.221	

Standard Error in parentheses.

*** Significant at 0.01.

respondents' income positively affects their choices while cost has a negative effect (Jin et al., 2006; Kanchanaroek et al., 2013; Vecchiato and Tempesta, 2013). These are encouraging results in terms of consistency with economic theory (Bateman et al., 2002; Riera et al., 2012). The scope test is another form of validation in stated preference-based methods. To test for the scope effect, we estimate a logistic regression where the dependent variable is binary indicating whether or not an alternative is chosen by the participant. The explanatory variables are the cost of the alternative and dichotomous variables indicating the number of recreation improvements projects in the alternative relative to the status quo. All recreation improvement coefficients in the logistic regression are significant and positive, suggesting that, *ceteris paribus*, greater number of improvements in recreation opportunities increases the probability of selecting an alternative (Table 7). In other words, consistent with the scope test, a greater number of improvements is preferred to fewer improvements. Although this form of the scope test disregards the magnitude of improvements it offers a reasonable first order assessment of the scope effect. In this study, we do not address other validity checks, such as transitivity or adding up tests (Hoyos, 2010; Elbakidze and Nayga, 2018) because our experimental design does not provide a convenient opportunity to include these tests. However, we control for comprehension and protest behavior of the respondents to ensure that the data used in the econometric analysis is reliable.

The respondents were told that the community garden could be used to generate income to be reinvested in the forest. However, no information was provided about how the garden would be managed. We do not investigate the effects of different garden maintenance or money collection scenarios. Similarly, we do not address maintenance or other specifics for the other recreation projects (e.g. width of the trail, quality of binoculars, features of the stage, etc.). These factors are relevant and should be addressed in future studies. Although lack of information about the specifics of the recreation enhancement projects may raise uncertainties about project outcomes, with possible implications for the responses and quality of econometric estimation (Munro and Hanley, 2001; Roberts et al., 2008; Wielgus et al., 2009), survey and focus group respondents in our project have not raised these concerns or asked for clarifications. Although presenting exact and exhaustive characterization of alternatives and attributes is often infeasible in the choice experiments, it is generally expected that better description of options enables better informed participant choices, which improves quality of econometric estimates.

5. Conclusions

Urban forests provide a bundle of benefits, including recreation opportunities that contribute to residents' quality of life. Nevertheless, the economic value of urban forest-based recreation has not been widely studied, particularly in the Caribbean region. We use a choice experiment to examine residents' preferences and WTP for different

recreation enhancement projects in the San Patricio Urban Forest in San Juan, Puerto Rico. Our results show that residents' WTP values for improvements in recreation opportunities, including trail enhancements, an event stage, binocular stands, and a community garden, vary between \$15 and \$39 depending on the project and on the estimation model. The results across the estimation models consistently show that among the considered enhancements the residents value community gardens the most, followed by the improvements to trails. Installation of an event stage is least preferred. One possible explanation for the low value of an event stage installation is that the residents may prefer preserving the relatively undisturbed condition of the urban forest relative to transforming the forest into a park with community events.

The results of this study shed light on the residents' preferences for recreation opportunities in an urban forest in San Juan. These results can be used as part of cost benefit analysis for individual project as well as for comparing relative economic merits of proposed projects. Future urban forest development project should reflect project costs as well as preferences of the residents. For example, investing in binocular stands or a stage rather than space for community garden or managed trails would be economically suboptimal if costs are comparable.

Acknowledgments

This work was supported by the USDA National Institute of Food and Agriculture, McIntire Stennis project 1012425. We thank Professor Luis Méndez from the University of Puerto Rico and Magaly Figueroa from US Forest Service for their support of this project. We thank three graduate students from the University of Puerto Rico for assistance with the focus groups, data collection and maps: María Méndez, Mónica Flores and Carlos Pérez. We thank Doreen Parés for assistance with focus groups.

Declaration of Competing Interest

None.

References

- Abildtrup, J., García, S., Olsen, S.B., Stenger, A., 2013. Spatial preference heterogeneity in forest recreation. *Ecol. Econ.* 92, 67–77.
- Adams, D.C., Bwenge, A.N., Lee, D.J., Larkin, S.L., Alavalapati, J.R., 2011. Public preferences for controlling upland invasive plants in state parks: application of a choice model. *Forest Policy Econ.* 13 (6), 465–472.
- Alpizar, F., Carlsson, F., Martinsson, P., 2003. Using choice experiments for non-market valuation. *Econ. Issues* 8, 83–110.
- Barrio, M., Loureiro, M.L., 2010. A meta-analysis of contingent valuation forest studies. *Ecol. Econ.* 69 (5), 1023–1030.
- Bateman, I.J., Carson, R.T., Day, B., Hanemann, W.M., Hanley, N., Hett, T., Jones-Lee, M., Loomes, G., Mourato, S., Ozdemiroglu, E., Pearce, D.W., Sugden, R., Swanson, S., 2002. *Economic Valuation With Stated Preference Techniques: A Manual*. Edward Elgar, Massachusetts, USA.
- Biol, E., Karousakis, K., Koundouri, P., 2006. Using economic valuation techniques to inform water resources management: a survey and critical appraisal of available techniques and an application. *Sci. Total Environ.* 365, 105–122.
- Bolund, P., Hunhammar, S., 1999. Ecosystem services in urban areas. *Ecol. Econ.* 29, 293–301.
- Boncinelli, F., Riccioli, F., Marone, E., 2015. Do forests help to keep my body mass index low? *Forest Policy Econ.* 54, 11–17.
- Brey, R., Riera, P., Mogas, J., 2007. Estimation of forest values using choice modeling: an application to Spanish forests. *Ecol. Econ.* 64 (2), 305–312.
- Champ, P.A., Boyle, K.J., Brown, T.C., 2003. *A Primer on Nonmarket Valuation*. Kluwer Academic Publishers, Norwell MA.
- Chiabai, A., Traversi, C.M., Markandya, A., Ding, H., Nunes, P.A., 2011. Economic assessment of forest ecosystem services losses: cost of policy inaction. *Environ. Resour. Econ.* 50 (3), 405–445.
- Chiesura, A., 2004. The role of urban parks for the sustainable city. *Landsc. Urban Plan.* 68, 129–138.
- Christie, M., Hanley, N., Hynes, S., 2007. Valuing enhancements to forest recreation using choice experiment and contingent behaviour methods. *J. For. Econ.* 13, 75–102.
- CPBSP (Ciudadanos Pro Bosque San Patricio), 2003. *Su Historia y su gente: Bosque San Patricio*. Retrieved from: http://www.bosquesanpatricio.org/files/documentos/Bosque_San_Patricio_Historial_del_Terreno.pdf.
- Czajkowski, M., Bartczak, A., Budziński, W., Giergiczny, M., Hanley, N., 2016. Preference and WTP stability for public forest management. *Forest Policy Econ.* 71, 11–22.

- de Groot, R.S., Wilson, M.A., Boumans, R.M.J., 2002. A typology for the classification, description and valuation of ecosystem functions, goods and services. *Ecol. Econ.* 41, 393–408.
- Domínguez-Torreiro, M., Soliño, M., 2011. Provided and perceived status quo in choice experiments: Implications for valuing the outputs of multifunctional rural areas. *Ecol. Econ.* 70 (12), 2523–2531.
- DRNA (Puerto Rico Department of Natural and Environmental Resources), 2008. Bosque de Puerto Rico: Bosque de San Patricio. Hojas de Nuestro Ambiente. Report P-026. Retrieved from: http://www.bosquesanpatricio.org/files/documentos/Boletin_DRNA_Bosque_SP.pdf.
- Dwyer, J.F., McPherson, E.G., Schroeder, H.W., Rowntree, R.A., 1992. Assessing the benefits and costs of the urban forest. *J. Arboric.* 18 (5), 227–234.
- Elbakidze, L., Nayga, R.M., 2018. The adding-up test in an incentivized value elicitation mechanism: The role of the income effect. *Environ. Resour. Econ.* 71 (3), 625–644.
- Escobedo, F.J., Kroeger, T., Wagner, J.E., 2011. Urban forests and pollution mitigation: Analyzing ecosystem services and disservices. *Environ. Pollut.* 159, 2078–2087.
- Ferraro, P.J., Lawlor, K., Mullan, K.L., Pattanayak, S.K., 2011. Forest figures: Ecosystem services valuation and policy evaluation in developing countries. *Rev. Environ. Econ. Policy* 6 (1), 20–44.
- Giergiczny, M., Czajkowski, M., Żylicz, T., Angelstam, P., 2015. Choice experiment assessment of public preferences for forest structural attributes. *Ecol. Econ.* 119, 8–23.
- Gill, D.A., Schuhmann, P.W., Oxenford, H.A., 2015. Recreational diver preferences for reef fish attributes: economic implications of future change. *Ecol. Econ.* 111, 48–57.
- Gómez-Baggethun, E., Barton, D., 2013. Classifying and valuing ecosystem services for urban planning. *Ecol. Econ.* 86, 235–245.
- Greene, W.H., 2012. *Econometric Analysis*, 7th ed. Prentice Hall, NJ, USA.
- Gürlük, S., 2006. The estimation of ecosystem services' value in the region of Misi Rural Development Project: Results from a contingent valuation survey. *Forest Policy Econ.* 9 (3), 209–218.
- Hanley, N., Wright, R.E., Adamowicz, V., 1998. Using choice experiments to value the environment. *Environ. Resour. Econ.* 11 (3–4), 413–428.
- Hausman, J., McFadden, D., 1984. Specification tests for the multinomial logit model. *Econometrica* 52 (5), 1219–1240.
- Hensher, D., 2006. Revealing differences in willingness to pay due to the dimensionality of stated choice designs: An initial assessment. *Environ. Resour. Econ.* 34, 7–44.
- Hoyos, D., 2010. The state of the art of environmental valuation with discrete choice experiments. *Ecol. Econ.* 69, 1595–1603.
- Irvine, S., Johnson, L., Peters, K., 1999. Community gardens and sustainable land use planning: a case study of the Alex Wilson Community Garden. *Local Environ.* 4, 33–46.
- Jacobson, M., Shr, Y.H., Dalemans, F., Magaju, C., Ciannella, R., 2018. Using a choice experiment approach to assess production tradeoffs for developing the croton value chain in Kenya. *Forest Policy Econ.* 86, 76–85.
- Japelj, A., Mavsar, R., Hodges, D., Kovač, M., Juvančič, L., 2016. Latent preferences of residents regarding an urban forest recreation setting in Ljubljana, Slovenia. *Forest Policy Econ.* 71, 71–79.
- Jim, C.Y., Chen, W.Y., 2008. Assessing the ecosystem service of air pollutant removal by urban trees in Guangzhou (China). *J. Environ. Manag.* 88, 665–676.
- Jin, J., Wang, Z., Ran, S., 2006. Comparison of contingent valuation and choice experiment in solid waste management programs in Macao. *Ecol. Econ.* 57, 430–441.
- Johnston, R.J., Boyle, K.J., Adamowicz, W., Bennett, J., Brouwer, R., Cameron, T.A., ..., Tourangeau, R., 2017. Contemporary guidance for stated preference studies. *J. Assoc. Environ. Resour. Econ.* 4 (2), 319–405.
- Juutinen, A., Kosenius, A., Ovaskainen, 2014. Estimating the benefits of recreation-oriented management in state-owned commercial forests in Finland: A choice experiment. *J. For. Econ.* 20, 396–412.
- Kanchanaroeck, Y., Termansen, M., Quinn, C., 2013. Property rights regimes in complex fishery management systems: A choice experiment application. *Ecol. Econ.* 93, 363–373.
- Lancaster, K., 1966. A new approach to consumer theory. *J. Polit. Econ.* 74, 132–157.
- Louvière, J., Hensher, D., Swait, J., 2000. *Stated Choice Methods. Analysis and Application*. Cambridge University Press, Cambridge.
- Majumdar, S., Deng, J., Zhang, Y., Pierskalla, C., 2011. Using contingent valuation to estimate the willingness of tourists to pay for urban forests: a study in Savannah, Georgia. *Urban Forest. Urban Green.* 10, 275–280.
- Malone, T., Lusk, J.L., 2018. Consequences of participant inattention with an application to carbon taxes for meat products. *Ecol. Econ.* 145, 218–230.
- Malone, T., Lusk, J.L., 2019. Releasing the trap: a method to reduce inattention bias in survey data with application to us beer taxes. *Econ. Inq.* 57 (1), 584–599.
- McFadden, D., 1974. Conditional logit analysis of qualitative choice behavior. In: Zarembka, P. (Ed.), *Frontiers in Econometrics*. Academic Press, New York.
- MEA (Millennium Ecosystem Assessment), 2005. *Ecosystems And Human Well-being: Synthesis*. Island Press, Washington, DC.
- Meyerhoff, J., Liebe, U., 2009. Status quo effect in choice experiments: empirical evidence on attitudes and choice task complexity. *Land Econ.* 85, 515–528.
- Munro, A., Hanley, N.D., 2001. Information, uncertainty and contingent valuation. In: Bateman, I., Willis, K.G. (Eds.), *Valuing Environmental Preferences: Theory and Practice of the Contingent Valuation Method in the US, EU, and Developing Countries*. Oxford University Press, Oxford, New York, United States.
- Ninan, K.N., Inoue, M., 2013. Valuing forest ecosystem services: what we know and what we don't. *Ecol. Econ.* 93, 137–149.
- Olschewski, R., Bebi, P., Teich, M., Hayek, U.W., Grêt-Regamey, A., 2012. Avalanche protection by forests — a choice experiment in the Swiss Alps. *Forest Policy Econ.* 17, 19–24.
- Park, T., Loomis, J.B., Creel, M., 1991. Confidence intervals for evaluating benefits from dichotomous choice contingent valuation studies. *Land Econ.* 67, 64–73.
- Pattanayak, S.K., Butry, D.T., 2005. Spatial complementarity of forests and farms: accounting for ecosystem services. *Am. J. Agric. Econ.* 87 (4), 995–1008.
- Penn, J., Hu, W., Cox, L., Kozloff, L., 2016. Values for recreational beach quality in Oahu, Hawaii. *Mar. Resour. Econ.* 31 (1), 47–62.
- Puerto Rico Planning Board, 2017. Socio-economic data, by municipality. Puerto Rico Government, San Juan, Puerto Rico.
- Richards, T.J., Hamilton, S.F., Allender, W.J., 2014. Social networks and new product choice. *Am. J. Agric. Econ.* 96 (2), 489–516.
- Ricketts, T.H., Daily, G.C., Ehrlich, P.R., Michener, C.D., 2004. Economic value of tropical forest to coffee production. *PNAS* 101, 12579–12582.
- Riera, P., Signorello, G., Thiene, M., Mahieu, P.A., Navrud, S., Kaval, P., ... Dragoi, S., 2012. Non-market valuation of forest goods and services: Good practice guidelines. *J. For. Econ.* 18, 259–270.
- Roberts, D.C., Boyer, T.A., Lusk, J.L., 2008. Preferences for environmental quality under uncertainty. *Ecol. Econ.* 66 (4), 584–593.
- Rolfe, J., Bennett, J., Louvière, J., 2000. Choice modelling and its potential application to tropical rainforest preservation. *Ecol. Econ.* 35 (2), 289–302.
- Rosenberger, R.S., Needham, M.D., Morzillo, A.T., Moehrke, C., 2012. Attitudes, willingness to pay, and stated values for recreation use fees at an urban proximate forest. *J. For. Econ.* 18, 271–281.
- Saldívar-Tanaka, L., Krasny, M.E., 2004. Culturing community development, neighborhood open space, and civic agriculture: the case of Latino community gardens in New York City. *Agric. Hum. Values* 21, 399–412.
- Schuhmann, P.W., Casey, J.F., Horrocks, J.A., Oxenford, H.A., 2013. Recreational SCUBA divers' willingness to pay for marine biodiversity in Barbados. *J. Environ. Manag.* 121, 29–36.
- Sheremet, O., Ruokamo, E., Juutinen, A., Svento, R., Hanley, N., 2018. Incentivising participation and spatial coordination in payment for ecosystem service schemes: forest disease control programs in Finland. *Ecol. Econ.* 152, 260–272.
- Termansen, M., McClean, C.J., Jensen, F.S., 2013. Modeling and mapping spatial heterogeneity in forest recreation services. *Ecol. Econ.* 92, 48–57.
- Train, K.E., 2009. *Discrete Choice Methods With Simulations*, 2nd edition. Cambridge University Press, NY, USA.
- Tyrväinen, L., Mäntymaa, E., Ovaskainen, V., 2014. Demand for enhanced forest amenities in private lands: The case of the Ruka-Kuusamo tourism area, Finland. *Forest Policy Econ.* 47, 4–13.
- Upton, V., Dhubbain, A.N., Bullock, C., 2012. preferences and values for afforestation: the effects of location and respondent understanding on forest attributes in a labeled choice experiment. *Forest Policy Econ.* 23, 17–27.
- U.S. Census Bureau, 2018. QuickFacts.
- Vecchiato, D., Tempesta, T., 2013. Valuing the benefits of an afforestation project in a peri-urban area with choice experiments. *Forest Policy Econ.* 26, 111–120.
- Vujčić, M., Tomicevic-Dubljevic, J., 2018. Urban forest benefits to the younger population: The case study of the city of Belgrade, Serbia. *Forest Policy Econ.* 96, 54–62.
- Wielgus, J., Gerber, L.R., Sala, E., Bennett, J., 2009. Including risk in stated-preference economic valuations: Experiments of choices for marine recreation. *J. Environ. Manag.* 90 (11), 3401–3409.
- Wuepper, D., Clemm, A., Wree, P., 2019. The preference for sustainable coffee and a new approach for dealing with hypothetical bias. *J. Econ. Behav. Organ.* 158, 475–486.